

Safety Data Sheet

Vitamin A-Palmitate 1.0 Mio IU/G stabilized with Tocopherol

Revision date : 2020/06/05

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Version: 4.0

(30041043/SDS_GEN_US/EN)

1. Identification

Product identifier used on the label

Vitamin A-Palmitate 1.0 Mio IU/G stabilized with Tocopherol

Recommended use of the chemical and restriction on use

Recommended use*: food additive(s)

* The "Recommended use" identified for this product is provided solely to comply with a Federal requirement and is not part of the seller's published specification. The terms of this Safety Data Sheet (SDS) do not create or infer any warranty, express or implied, including by incorporation into or reference in the seller's sales agreement.

Details of the supplier of the safety data sheet

Company:
BASF CORPORATION
100 Park Avenue
Florham Park, NJ 07932, USA

Telephone: +1 973 245-6000

Emergency telephone number

CHEMTREC: 1-800-424-9300
BASF HOTLINE: 1-800-832-HELP (4357)

Other means of identification

Synonyms: Preparation based on: Retinyl palmitate

2. Hazards Identification

According to Regulation 2012 OSHA Hazard Communication Standard; 29 CFR Part 1910.1200

Classification of the product

Skin Sens.	1	Skin sensitization
Repr.	1B (unborn child)	Reproductive toxicity
Aquatic Chronic	4	Hazardous to the aquatic environment - chronic

Label elements

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Pictogram:



Signal Word:

Danger

Hazard Statement:

H317	May cause an allergic skin reaction.
H360	May damage the unborn child.
H413	May cause long lasting harmful effects to aquatic life.

Precautionary Statements (Prevention):

P280	Wear protective gloves, protective clothing and eye protection or face protection.
P261	Avoid breathing dust/fume/gas/mist/vapours/spray.
P201	Obtain special instructions before use.
P273	Avoid release to the environment.
P202	Do not handle until all safety precautions have been read and understood.
P272	Contaminated work clothing should not be allowed out of the workplace.

Precautionary Statements (Response):

P303 + P352	IF ON SKIN (or hair): Wash with plenty of soap and water.
P308 + P311	IF exposed or concerned: Call a POISON CENTER or physician.
P362 + P364	Take off contaminated clothing and wash it before reuse.

Precautionary Statements (Storage):

P405	Store locked up.
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Precautionary Statements (Disposal):

P501	Dispose of contents and container to hazardous or special waste collection point.
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Hazards not otherwise classified

When finely distributed, self-ignition is possible.

3. Composition / Information on Ingredients

According to Regulation 2012 OSHA Hazard Communication Standard; 29 CFR Part 1910.1200

D,L-alpha-Tocopherol

CAS Number: 10191-41-0

Content (W/W): 1.0 - 3.0%

Synonym: 2H-1-Benzopyran-6-ol, 3,4-dihydro-2,5,7,8-tetramethyl-2-(4,8,12-trimethyltridecyl)-

Sunflower oil

CAS Number: 8001-21-6

Content (W/W): 30.0 - 50.0%

Synonym: No data available.

Vitamin A palmitate

CAS Number: 79-81-2

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Content (W/W): 40.0 - 60.0%

Synonym: No data available.

4. First-Aid Measures

Description of first aid measures

General advice:

First aid personnel should pay attention to their own safety. If the patient is likely to become unconscious, place and transport in stable sideways position (recovery position). Immediately remove contaminated clothing.

If inhaled:

Keep patient calm, remove to fresh air, seek medical attention.

If on skin:

Immediately wash thoroughly with soap and water, seek medical attention.

If in eyes:

Wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist.

If swallowed:

Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.

Most important symptoms and effects, both acute and delayed

Information on: Vitamin A palmitate

Symptoms: Overexposure may cause: Eye irritation, skin irritation, erythema, nausea, headache, vomiting, dizziness, diarrhea, abdominal cramps

Indication of any immediate medical attention and special treatment needed

Note to physician

Treatment:

Treat according to symptoms (decontamination, vital functions), no known specific antidote.

5. Fire-Fighting Measures

Extinguishing media

Suitable extinguishing media:

water spray, carbon dioxide, dry powder, foam

Unsuitable extinguishing media for safety reasons:

water jet

Special hazards arising from the substance or mixture

Hazards during fire-fighting:

2-propenal, carbon oxides, harmful vapours

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The substances/groups of substances mentioned can be released in case of fire. Evolution of fumes/fog.

Advice for fire-fighters

Protective equipment for fire-fighting:

Firefighters should be equipped with self-contained breathing apparatus and turn-out gear.

Further information:

In case of combustion evolution of toxic gases/vapours possible. Cool endangered containers with water-spray. Collect contaminated extinguishing water separately, do not allow to reach sewage or effluent systems. Dispose of fire debris and contaminated extinguishing water in accordance with official regulations.

6. Accidental release measures

Personal precautions, protective equipment and emergency procedures

Use personal protective clothing. Information regarding personal protective measures, see section 8. Ensure adequate ventilation. Do not breathe vapour/aerosol/spray mists. Avoid contact with the skin, eyes and clothing.

Environmental precautions

Do not discharge into drains/surface waters/groundwater. Inform authorities in the event of product spillage to water courses or sewage systems.

May be harmful to the aquatic environment. Prevent entry into drains and surface waters.

Methods and material for containment and cleaning up

For small amounts: Pick up with suitable absorbent material. After taking up material in containers, cover immediately with water layer.

For large amounts: Dike spillage. Pump off product.

Dispose of absorbed material in accordance with regulations. Mop up spills with non-flammable adsorbents (e.g. vermiculite, spill mats). Soiled textiles / cleaning rags / adsorbents and Silica are capable of self ignition and should be wetted with water and must be disposed of in a safe manner.

7. Handling and Storage

Precautions for safe handling

Avoid aerosol formation. Wear suitable protective clothing and eye/face protection. Avoid contact with the skin, eyes and clothing. Keep container tightly sealed. Ensure that there is no crystallized product in the container before use. Processing machines must be fitted with local exhaust ventilation.

Protection against fire and explosion:

Risk of self-ignition when a large surface area is produced due to fine dispersion. Soiled textiles / cleaning rags / adsorbents and Silica are capable of self ignition and should be wetted with water and must be disposed of in a safe manner. Take precautionary measures against static discharges. Avoid all sources of ignition: heat, sparks, open flame. If exposed to fire, keep containers cool by spraying with water.

Conditions for safe storage, including any incompatibilities

Segregate from oxidants.

Suitable materials for containers: High density polyethylene (HDPE), Low density polyethylene (LDPE)

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Further information on storage conditions: Keep container tightly closed and dry; store in a cool place. Protect from air. Protect from the effects of light. Keep under nitrogen.

8. Exposure Controls/Personal Protection

Components with occupational exposure limits

Sunflower oil	OSHA PEL	PEL 15 mg/m3 Total dust ; PEL 5 mg/m3 Respirable fraction ; TWA value 5 mg/m3 Respirable fraction ; TWA value 15 mg/m3 Total dust ;
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Personal protective equipment

Respiratory protection:

Wear a NIOSH-certified (or equivalent) organic vapour/particulate respirator as needed.

Hand protection:

Wear chemical resistant protective gloves.

Eye protection:

Tightly fitting safety goggles (chemical goggles).

Body protection:

Body protection must be chosen depending on activity and possible exposure, e.g. head protection, apron, protective boots, chemical-protection suit.

General safety and hygiene measures:

Handle in accordance with good industrial hygiene and safety practice. Wearing of closed work clothing is required additionally to the stated personal protection equipment. Under no circumstances should the product come into contact with the skin of pregnant women or be inhaled by them. Avoid contact with skin. No eating, drinking, smoking or tobacco use at the place of work. Hands and/or face should be washed before breaks and at the end of the shift. Store work clothing separately.

9. Physical and Chemical Properties

Form:	oily	
Odour:	not applicable	
Odour threshold:	Not determined due to potential health hazard by inhalation.	
Colour:	yellow	
pH value:	insoluble	
Freezing point:	not determined	
Melting point:	not determined	
Boiling point:	The substance / product decomposes therefore not determined.	
Flash point:	> 100 °C	(DIN 51758)
Flammability:	hardly combustible	(derived from flash point)
Lower explosion limit:	For liquids not relevant for classification and labelling.	
Upper explosion limit:	For liquids not relevant for classification and labelling.	
Autoignition:	approx. 265 °C	(DIN 51794)

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Vapour pressure:	negligible
Density:	0.88 g/cm ³ (20 °C)
Vapour density:	negligible
Partitioning coefficient n-octanol/water (log Pow):	not applicable for mixtures
Self-ignition temperature:	Risk of self-ignition when a large surface area is produced due to fine dispersion.
Thermal decomposition:	No decomposition if stored and handled as prescribed/indicated.
Viscosity, dynamic:	27 mPa.s (60 °C)
Viscosity, kinematic:	No data available.
Particle size:	The substance / product is marketed or used in a non solid or granular form.
Solubility in water:	insoluble
Solubility (qualitative):	soluble solvent(s): hydrocarbons, chlorinated hydrocarbons, ether, fats, oils,
Molar mass:	No data available.
Evaporation rate:	negligible

10. Stability and Reactivity

Reactivity

Corrosion to metals:
Corrosive effects to metal are not anticipated.

Oxidizing properties:
Based on its structural properties the product is not classified as oxidizing.

Chemical stability

The product is stable if stored and handled as prescribed/indicated.

Possibility of hazardous reactions

When finely distributed, self-ignition is possible.

Conditions to avoid

Avoid light. Avoid all sources of ignition: heat, sparks, open flame. Avoid electro-static charge. See SDS section 7 - Handling and storage.

Incompatible materials

atmospheric oxygen, atmospheric moisture

Hazardous decomposition products

Decomposition products:
Hazardous decomposition products: No hazardous decomposition products if stored and handled as prescribed/indicated.

Thermal decomposition:
No decomposition if stored and handled as prescribed/indicated.

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11. Toxicological information

Primary routes of exposure

Routes of entry for solids and liquids are ingestion and inhalation, but may include eye or skin contact. Routes of entry for gases include inhalation and eye contact. Skin contact may be a route of entry for liquefied gases.

Acute Toxicity/Effects

Acute toxicity

Assessment of acute toxicity: Virtually nontoxic after a single ingestion.

Oral

Information on: Vitamin A palmitate

Type of value: LD50

Species: rat (male/female)

Value: > 2,000 mg/kg (BASF-Test)

No mortality was observed. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Inhalation

No data available.

Dermal

No data available.

Assessment other acute effects

Assessment of STOT single:

Based on available Data, the classification criteria are not met.

Irritation / corrosion

Assessment of irritating effects: Skin contact causes slight irritation. May cause slight irritation to the eyes.

Information on: Vitamin A palmitate

Assessment of irritating effects: Not irritating to the eyes. May cause slight irritation to the skin.

Information on: Sunflower oil

Assessment of irritating effects: May cause slight irritation to the skin. May cause slight irritation to the eyes. May cause slight irritation to the respiratory tract. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Information on: D,L-alpha-Tocopherol

Assessment of irritating effects: Not irritating to the skin. Not irritating to the eyes.

Skin

Species: rabbit

Method: BASF-Test

No data available.

Eye

Species: rabbit

Method: BASF-Test

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No data available.

Sensitization

Assessment of sensitization: Sensitization after skin contact possible.

Information on: D,L-alpha-Tocopherol

Assessment of sensitization:

Caused skin sensitization in animal studies.

Aspiration Hazard

No data available.

Chronic Toxicity/Effects

Repeated dose toxicity

Assessment of repeated dose toxicity: Based on available Data, the classification criteria are not met.

Information on: Vitamin A palmitate

Assessment of repeated dose toxicity: Repeated exposure to large quantities may affect certain organs.

Information on: D,L-alpha-Tocopherol

Assessment of repeated dose toxicity: Repeated oral uptake of the substance did not cause substance-related effects.

Genetic toxicity

Assessment of mutagenicity: Based on available Data, the classification criteria are not met.

Information on: Vitamin A palmitate

Assessment of mutagenicity: In the majority of tests performed (bacteria/microorganisms/cell cultures) a mutagenic effect was not found. A mutagenic effect was also not observed in in-vivo assays. The product has not been fully tested. The statements have been derived in parts from products of a similar structure or composition.

Information on: D,L-alpha-Tocopherol

Assessment of mutagenicity: Results from a number of mutagenicity studies with microorganisms, mammalian cell culture and mammals are available. Taking into account all of the information, there is no indication that the substance is mutagenic.

Carcinogenicity

Assessment of carcinogenicity: Based on available Data, the classification criteria are not met.

Information on: Vitamin A palmitate

Assessment of carcinogenicity: Results from a number of long-term carcinogenity studies and short-term tests are available. Taking into account all of the information, there is no indication that the substance itself is carcinogenic. Literature data.

Information on: D,L-alpha-Tocopherol

Assessment of carcinogenicity: In long-term studies in rats in which the substance was given by feed, a carcinogenic effect was not observed. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

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Reproductive toxicity

Assessment of reproduction toxicity: Not classified, due to lack of data.

Teratogenicity

Assessment of teratogenicity: The substance caused malformations/developmental toxicity in laboratory animals.

Information on: Vitamin A palmitate

Assessment of teratogenicity: May cause harm to the unborn child.

Other Information

The product has not been tested. The statements on toxicology have been derived from the properties of the individual components.

12. Ecological Information

Toxicity

Aquatic toxicity

Assessment of aquatic toxicity:

There is a high probability that the product is not acutely harmful to aquatic organisms. The inhibition of the degradation activity of activated sludge is not anticipated when introduced to biological treatment plants in appropriate low concentrations.

Toxicity to fish

Information on: Vitamin A palmitate

LC50 (96 h) > 10,000 mg/l, Leuciscus idus (DIN 38412 Part 15, static)

The details of the toxic effect relate to the nominal concentration. The product has not been tested.

The statement has been derived from substances/products of a similar structure or composition.

Information on: Sunflower oil

LC50 (96 h) > 1,000 mg/l, Brachydanio rerio (other)

The details of the toxic effect relate to the nominal concentration.

Aquatic invertebrates

Information on: Vitamin A palmitate

EC50 (48 h) > 100 mg/l, Daphnia magna (Screening test, static)

The details of the toxic effect relate to the nominal concentration. The product has not been tested.

The statement has been derived from substances/products of a similar structure or composition.

Information on: Sunflower oil

EC50 (48 h) > 100 mg/l, Daphnia magna (other)

The details of the toxic effect relate to the nominal concentration.

Aquatic plants

Information on: Vitamin A palmitate

EC50 (72 h) 152.94 mg/l (growth rate), Scenedesmus subspicatus (DIN 38412 Part 9, static)

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The product has low solubility in the test medium. An aqueous solution prepared with solubilizers has been tested.

Information on: Sunflower oil

EC50 > 100 mg/l, algae (other)

The details of the toxic effect relate to the nominal concentration.

Assessment of terrestrial toxicity

No data available.

Microorganisms/Effect on activated sludge

Toxicity to microorganisms

Information on: Vitamin A palmitate

*DIN EN ISO 8192-OECD 209-88/302/EEC,P. C aerobic
activated sludge, domestic/EC20 (30 min): > 1,000 mg/l*

Persistence and degradability

Assessment biodegradation and elimination (H2O)

Information on: Vitamin A palmitate

Not readily biodegradable (by OECD criteria). Moderately/partially biodegradable.

Information on: D,L-alpha-Tocopherol

Not readily biodegradable (by OECD criteria). Biodegradable.

Bioaccumulative potential

Assessment bioaccumulation potential

Information on: Vitamin A palmitate

The product will not be readily bioavailable due to its consistency and insolubility in water. No significant accumulation in organisms is expected as a result of the distribution coefficient of n-octanol/water (log Pow).

Bioaccumulation potential

Information on: Sunflower oil

Does not accumulate in organisms.

Mobility in soil

Assessment transport between environmental compartments

Information on: Vitamin A palmitate

The substance will not evaporate into the atmosphere from the water surface.

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Adsorption to solid soil phase is expected.

Additional information

Other ecotoxicological advice:

The product has not been tested. The statements on ecotoxicology have been derived from the properties of the individual components.

13. Disposal considerations

Waste disposal of substance:

Dispose of in accordance with national, state and local regulations.

Container disposal:

Dispose of in accordance with national, state and local regulations.

14. Transport Information

Land transport

USDOT

Not classified as a dangerous good under transport regulations

Sea transport

IMDG

Not classified as a dangerous good under transport regulations

Air transport

IATA/ICAO

Not classified as a dangerous good under transport regulations

15. Regulatory Information

Federal Regulations

Registration status:

Chemical TSCA, US released / listed

Cosmetic TSCA, US released / exempt

Food TSCA, US released / exempt

EPCRA 311/312 (Hazard categories): Refer to SDS section 2 for GHS hazard classes applicable for this product.

State regulations

State RTK

PA

CAS Number

8001-21-6

Chemical name

Sunflower oil

Safe Drinking Water & Toxic Enforcement Act, CA Prop. 65:

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WARNING: This product can expose you to chemicals including RETINOL PALMITATE, which is known to the State of California to cause birth defects or other reproductive harm. For more information, go to www.P65Warnings.ca.gov.

NFPA Hazard codes:

Health: 2 Fire: 1 Reactivity: 0 Special:

HMIS III rating

Health: 2^a Flammability: 1 Physical hazard: 0

Assessment of the hazard classes according to UN GHS criteria (most recent version):

Skin Corr./Irrit.	3	Skin corrosion/irritation
Skin Sens.	1	Skin sensitization
Aquatic Chronic	4	Hazardous to the aquatic environment - chronic
Repr.	1B (unborn child)	Reproductive toxicity

16. Other Information

SDS Prepared by:

BASF NA Product Regulations

SDS Prepared on: 2020/06/05

We support worldwide Responsible Care® initiatives. We value the health and safety of our employees, customers, suppliers and neighbors, and the protection of the environment. Our commitment to Responsible Care is integral to conducting our business and operating our facilities in a safe and environmentally responsible fashion, supporting our customers and suppliers in ensuring the safe and environmentally sound handling of our products, and minimizing the impact of our operations on society and the environment during production, storage, transport, use and disposal of our products.

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